SUMMATIVE ASSESSMENT – 1 (SA-1) MODEL PAPER – SET 2

CLASS IX · MATHEMATICS · Semester-1 · AP SCERT 2026-27 · SSC 100-Mark Pattern

BLUEPRINT – SA-1 MODEL PAPER – SET 2

SUBJECT: MATHEMATICS	CLASS: IX	TIME: 3 Hrs 15 Min	MAX. MARKS: 100

Syllabus: Class IX Mathematics, Semester-1 (AP SCERT 2026-27) – 1. Number Systems, 2. Polynomials, 3. Coordinate Geometry, 4. Linear Equations in Two Variables, 5. Introduction to Euclid's Geometry, 6. Lines and Angles. Pattern: SSC 100-mark Summative Assessment blueprint with Academic Standards (Awareness 60%, Sensitivity 20%, Creativity 20%).

1. Weightage to Academic Standards (Ability)

ABILITY	AWARENESS (60%)	SENSITIVITY (20%)	CREATIVITY (20%)	TOTAL
MARKS	60	20	20	100

2. Weightage to Form of Questions

FORM OF QUESTION	Essay / Long Ans.	Short Answer	Very Short Ans.	Objective	TOTAL
No. of questions	5	8	8	12	33
Marks allotted	8	4	2	1	100
Estimated time	70 min	60 min	30 min	20 min	180 + 15 min

3. Ability-wise Distribution of Questions

ABILITY	1 Mark	2 Marks	4 Marks	8 Marks	Total Marks
Awareness	4	4	6	3	60
Sensitivity	4	2	1	1	20
Creativity	4	2	1	1	20
Total Questions	12	8	8	5	100

4. Chapter-wise Weightage

S.No	UNIT / CHAPTER	MARKS
1	Number Systems	1+1+2+2+4+4+8+(8) = 22 + (8)
2	Polynomials	1+1+2+4+4+8+(8) = 20 + (8)
3	Linear Equations in Two Variables	1+1+2+4+8+(8) = 16 + (8)
4	Coordinate Geometry	1+1+2+4+8 = 16
5	Introduction to Euclid's Geometry	1+1+2+4+(8) = 8 + (8)
6	Lines and Angles	1+1+2+2+4+8+(8) = 18 + (8)
	TOTAL	**100 + (40)**

Note: Marks within ( ) are the internal-choice (OR) questions of Section IV. The 100 marks are distributed over the six Semester-1 chapters; the ability-wise (60 : 20 : 20) and form-wise weightage follow the SSC blueprint pattern.

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5. Question-wise Chapter Distribution and Domain (Questions 1 to 33)

Q.No	Chapter / Topic	Domain	Marks
1	Number Systems	Awareness	1
2	Polynomials	Creativity	1
3	Linear Equations in Two Variables	Creativity	1
4	Coordinate Geometry	Creativity	1
5	Introduction to Euclid’s Geometry	Sensitivity	1
6	Lines and Angles	Awareness	1
7	Number Systems	Sensitivity	1
8	Polynomials	Sensitivity	1
9	Introduction to Euclid’s Geometry	Creativity	1
10	Coordinate Geometry	Awareness	1
11	Linear Equations in Two Variables	Awareness	1
12	Lines and Angles	Sensitivity	1
13	Number Systems	Creativity	2
14	Number Systems	Awareness	2
15	Polynomials	Awareness	2
16	Coordinate Geometry	Sensitivity	2
17	Linear Equations in Two Variables	Awareness	2
18	Introduction to Euclid’s Geometry	Sensitivity	2
19	Lines and Angles	Awareness	2
20	Lines and Angles	Creativity	2
21	Number Systems	Awareness	4
22	Number Systems	Awareness	4
23	Polynomials	Awareness	4
24	Polynomials	Awareness	4
25	Linear Equations in Two Variables	Sensitivity	4
26	Coordinate Geometry	Awareness	4
27	Introduction to Euclid’s Geometry	Awareness	4
28	Lines and Angles	Creativity	4
29a / 29b	(a) Number Systems (b) Introduction to Euclid’s Geometry	Creativity	8
30a / 30b	(a) Polynomials (b) Number Systems	Awareness	8
31a / 31b	(a) Linear Equations in Two Variables (b) Polynomials	Sensitivity	8
32a / 32b	(a) Lines and Angles (b) Linear Equations in Two Variables	Awareness	8
33a / 33b	(a) Coordinate Geometry (b) Lines and Angles	Awareness	8

ABILITIES AND THEIR OBJECTIVES AS PER PARAKH NORMS

I) AWARENESS: Knowledge / understanding of the construct / understanding of the process (resembles old objectives Knowledge, Understanding and Application).

II) SENSITIVITY: Conflict resolution / open-mindedness / collaboration (resembles old objective Analysis).

III) CREATIVITY: Generate innovative ideas / flexibility / fluency / exploration / combining ideas and concepts (resembles old objectives Evaluation and Creativity).

NOTE: Overall, the chapter-wise and ability-wise weightage is followed. There is no rigid rule that a question with a particular number must come from a particular chapter.

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SUMMATIVE ASSESSMENT – 1 : MODEL QUESTION PAPER – SET 2

MATHEMATICS – CLASS IX (Semester-1, AP SCERT Syllabus 2026-27)

Time : 3 Hrs 15 Min (15 min reading + 3 hrs writing) **Max. Marks : 100**

General Instructions: *(i) Answer all the questions. (ii) Write the answers neatly in the answer booklet. (iii) The paper has four Sections (I, II, III, IV). (iv) Use of calculators is not permitted. (v) Draw neat, labelled figures wherever necessary.*

SECTION – I 12 × 1 = 12 Marks

Note : *i) Answer all the questions. ii) Each question carries 1 mark. iii) Choose the correct option and write its letter (A/B/C/D) in the answer booklet.*

1. Which of the following is a rational number? *[Awareness · Number Systems]*

(A) $\sqrt{2}$ (B) π (C) 0.1010010001… (D) 0.121212…

2. If $(x + 1)$ is a factor of $x^3 + ax^2 - x + 2$, then a = *[Creativity · Polynomials]*

(A) 2 (B) −2 (C) 1 (D) −1

3. If $(k, -2)$ is a solution of $3x + 2y = 5$, then k = *[Creativity · Linear Equations in Two Variables]*

(A) 1 (B) 2 (C) 3 (D) 4

4. The signs of the coordinates of a point in the fourth quadrant are *[Creativity · Coordinate Geometry]*

(A) (+, +) (B) (−, +) (C) (−, −) (D) (+, −)

5. Which of the following statements is true? *[Sensitivity · Introduction to Euclid's Geometry]*

(A) A line segment has two end points (B) A line has two end points
(C) A ray has two end points (D) A point has length

6. The complement of 40° is *[Awareness · Lines and Angles]*

(A) 140° (B) 50° (C) 60° (D) 130°

7. $(\sqrt{3} + \sqrt{2})^2$ = *[Sensitivity · Number Systems]*

(A) 5 (B) $5 + 2\sqrt{6}$ (C) $5 - 2\sqrt{6}$ (D) 1

8. The remainder when $x^3 - 3x^2 + 2x$ is divided by $(x - 1)$ is *[Sensitivity · Polynomials]*

(A) 0 (B) 1 (C) 2 (D) −1

9. Euclid defined a line as 'breadthless length'. So a line has *[Creativity · Introduction to Euclid's Geometry]*

(A) length only (B) length and breadth (C) breadth only (D) none of these

10. The perpendicular distance of the point (4, 3) from the x-axis is *[Awareness · Coordinate Geometry]*

(A) 4 (B) 3 (C) 5 (D) 7

11. The equation of the x-axis is *[Awareness · Linear Equations in Two Variables]*

(A) $x = 0$ (B) $y = 0$ (C) $x + y = 0$ (D) $x = y$

12. In ΔABC, $\angle A = 40°$ and $\angle B = 70°$. The exterior angle at C is *[Sensitivity · Lines and Angles]*

(A) 70° (B) 110° (C) 140° (D) 40°

SECTION – II 8 × 2 = 16 Marks

Note : *i) Answer all the questions. ii) Each question carries 2 marks.*

13. Find two irrational numbers between 0.12 and 0.13. *[Creativity · Number Systems]*

14. Express 1.2727… in the form p/q. *[Awareness · Number Systems]*

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15. Find the zero of the polynomial p(x) = 2x + 5. *[Awareness · Polynomials]*

16. Rani says the points (3, −2) and (−2, 3) are the same, but Rama disagrees. Who is correct? Why? *[Sensitivity · Coordinate Geometry]*

17. Find three solutions of the equation 2x + y = 7. *[Awareness · Linear Equations in Two Variables]*

18. Rakesh claims that many different lines can be drawn through two distinct points. Using Euclid's axiom, explain whether he is right. *[Sensitivity · Introduction to Euclid's Geometry]*

19. If ∠AOC = (2x + 20)° and ∠BOC = (3x + 10)° form a linear pair, find x and the angles. *[Awareness · Lines and Angles]*

20. A transversal cuts two parallel lines. One of a pair of alternate interior angles is (2x + 15)° and the other is (3x − 10)°. Find x and the angles. *[Creativity · Lines and Angles]*

SECTION – III 8 × 4 = 32 Marks

Note : *i) Answer all the questions. ii) Each question carries 4 marks.*

21. Rationalise: (i) $1/(3 + 2\sqrt{2})$ (ii) $5/(\sqrt{3} - \sqrt{5})$. *[Awareness · Number Systems]*

22. Simplify: (i) $64^{-1/3}$ (ii) $32^{2/5}$ (iii) $5^{1/4} \times 5^{3/4}$ (iv) $7^{1/2} \times 11^{1/2}$. *[Awareness · Number Systems]*

23. Factorise: $x^3 + 13x^2 + 32x + 20$. *[Awareness · Polynomials]*

24. Expand: (i) $(x + 2y + 3z)^2$ (ii) $(3a - 2b)^3$. *[Awareness · Polynomials]*

25. A taxi charges Rs. 15 for the first km and Rs. 8 for each additional km. Taking the total distance as x km and the fare as Rs. y, write a linear equation. Meena says the fare for 3 km is Rs. 31 and Seema says it is Rs. 39. Who is correct? *[Sensitivity · Linear Equations in Two Variables]*

26. Plot the points A(−4, 0), B(0, 3), C(4, 0) and D(0, −3), join them in order and name the figure formed. *[Awareness · Coordinate Geometry]*

27. Write any five of Euclid's axioms. *[Awareness · Introduction to Euclid's Geometry]*

28. In ΔABC, side BC is produced to D. If ∠ACD = 110° and ∠A : ∠B = 2 : 3, find the angles of the triangle. *[Creativity · Lines and Angles]*

SECTION – IV 5 × 8 = 40 Marks

Note : *i) Answer all the questions. ii) Each question carries 8 marks. iii) There is an internal choice for each question.*

29.

a) i) Find six rational numbers between 2 and 3.
ii) Find three irrational numbers between 2/7 and 3/7. *[Creativity · Number Systems]*

OR

b) i) State Euclid's fifth postulate and its equivalent version (Playfair's axiom).
ii) If AC = BC, where C lies between A and B, prove that AC = ½AB. *[Creativity · Introduction to Euclid's Geometry]*

30.

a) Factorise $2x^3 - 5x^2 - 19x + 42$ using the factor theorem. *[Awareness · Polynomials]*

OR

b) If $x = 5 + 2\sqrt{6}$, find (i) $1/x$ (ii) $x + 1/x$ (iii) $x^2 + 1/x^2$. *[Awareness · Number Systems]*

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31.

a) Two pencils and one eraser cost Rs. 8 together. Write a linear equation in two variables (x = cost of a pencil, y = cost of an eraser), give four solutions in a table and draw its graph. From the graph find the cost of an eraser if a pencil costs Rs. 3. *[Sensitivity · Linear Equations in Two Variables]*

OR

b) Show that $(x - 2)$ and $(x + 3)$ are factors of $p(x) = x^3 - x^2 - 8x + 12$. Hence factorise $p(x)$ completely. *[Sensitivity · Polynomials]*

32.

a) i) Prove that the bisectors of the two angles forming a linear pair are at right angles.
ii) Two lines AB and CD intersect at O. If $\angle AOC = 4x°$ and $\angle BOC = 5x°$, find x and $\angle AOD$. *[Awareness · Lines and Angles]*

OR

b) Draw the graphs of $x = -2$, $y = 3$ and $2x + y = 5$ on the same graph paper and find the area of the triangle formed. *[Awareness · Linear Equations in Two Variables]*

33.

a) Plot the points P(−3, 2), Q(3, 2), R(3, −4) and S(−3, −4) and join them in order. (i) Name the figure. (ii) Find the length of each side. (iii) Find its area. (iv) Write the coordinates of the point where the diagonals meet. *[Awareness · Coordinate Geometry]*

OR

b) i) Prove that if a side of a triangle is produced, the exterior angle is equal to the sum of the two interior opposite angles.
ii) In ΔPQR, side QR is produced to S. If $\angle PQR = 40°$ and $\angle QPR = 65°$, find $\angle PRS$ and $\angle PRQ$. *[Awareness · Lines and Angles]*

— ALL THE BEST —

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